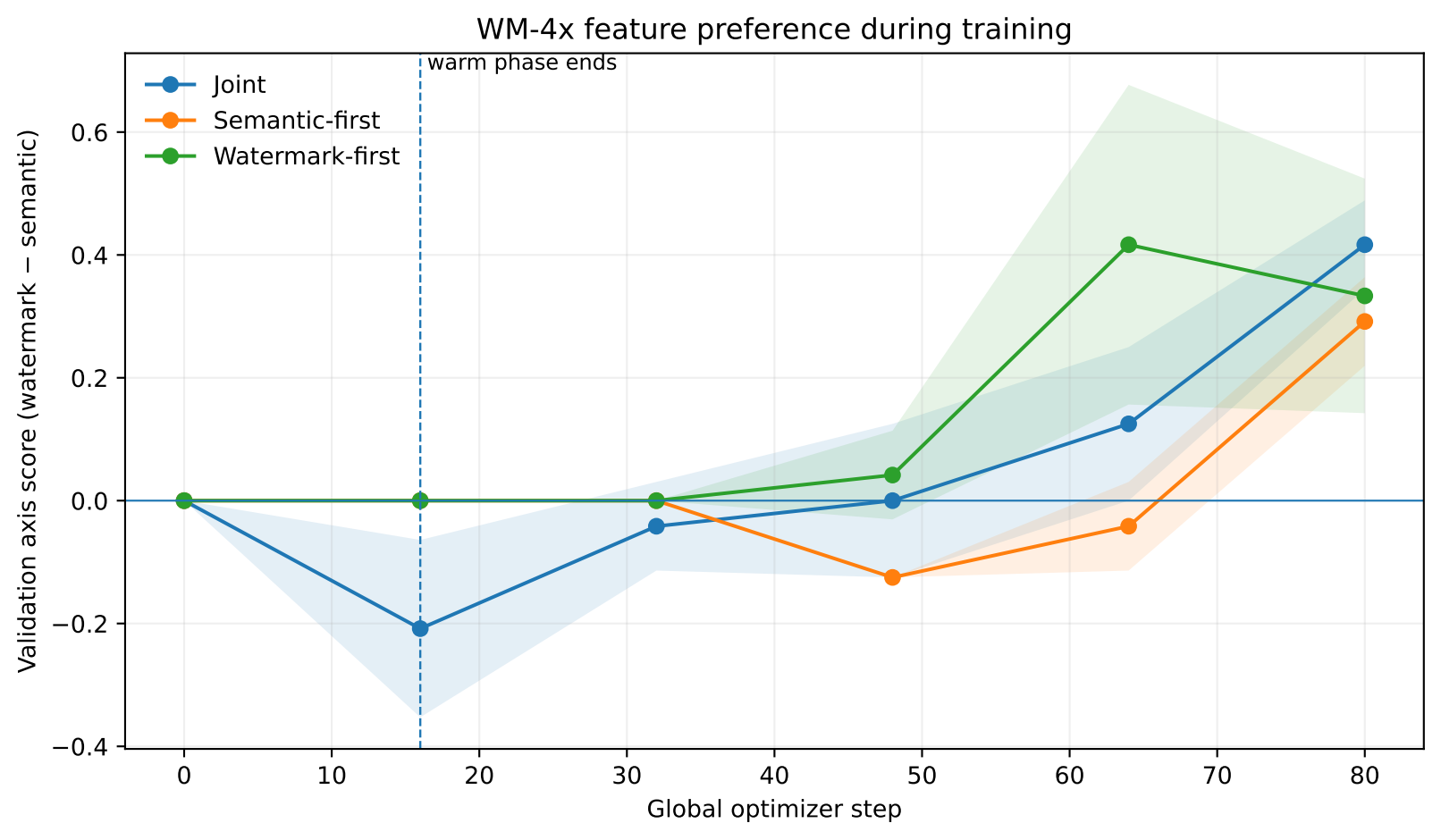
WM-4x feature preference during training
warm phase ends
Joint
Semantic-first
Watermark-first
Validation axis score (watermark − semantic)
0.6
0.4
0.2
0.0
−0.2
−0.4
0
10
20
30
40
50
60
70
80
Global optimizer step